Stokesov TM:

$$\oint_C z\,dx + 3\,dy - 2x\,dz \Rightarrow \iint_\Sigma (\mathrm{rot}\,\vec{f})\cdot\vec{n}\cdot d\vec{S} \rightarrow \text{3 etape ručnoj}$$

1) Rotacija f $= \mathrm{rot}\,\vec{f}$

2.) Normala $= \dfrac{\nabla}{\|\nabla\|} = \vec{n}$

3.) $dS = \sqrt{(z_x')^2 + (z_x')^2 + (z_y')^2}$

Green: ! po zatvorenoj plohi!

Teorem o divergenciji

$$\iint_\Sigma \vec{f}\cdot d\vec{S} = \iiint_V \mathrm{div}\,\vec{f}\cdot dV$$

Odusmo kad nije zatvorena ni je zatvorimo i onda oduzmemo te JEDNOSTAVNE plohe

$$\hookrightarrow \underbrace{\iint_\Sigma \vec{f}\cdot d\vec{S}}_{\text{ovaj tražimo}} + \underbrace{\iint_{\Sigma_0} \vec{f}\cdot d\vec{S}}_{\text{pomoćne}} = \iint_{\Sigma\cup\Sigma_0} \vec{f}\cdot d\vec{S} = \iiint_V \mathrm{div}\,\vec{f}\cdot dV$$

$$\boxed{\underbrace{\iint_\Sigma \vec{f}\cdot d\vec{S}}_{\text{(nezatvorena)}} = \underbrace{\iiint_V \mathrm{div}\,\vec{f}\cdot dV}_{\text{Greenov integ.}} - \iint_{\Sigma_0} \vec{f}\cdot d\vec{S}}$$

pomoćne plohe

normala $\uparrow$ $(+\vec{k})$

$$+\iint (-)\cdot z_{0_1}\,dx\,dy$$

$$-\iint (-)\cdot z_{0_1}\,dx\,dy$$

broj uz x u $\vec{f}$

TOK vekt polja kroz zadanu plohu: $\iint_\Sigma (\vec{f}\cdot\vec{n})\,dS$

$\iint_\Sigma (\vec{f}\cdot\vec{n})\,dS =$ 3 koraka opet

- $\hookrightarrow f$
- $\hookrightarrow \vec{n} \longrightarrow$ NA ZADANU PLOHU!
- $\hookrightarrow d\vec{S}$

KOMPLEKSNA ANALIZA:

$$z = x \pm iy$$

$$\cos(z) = \cos(x)\cdot\cosh(y) \overset{+}{-} i\sin(x)\cdot\sinh(y)$$

$$\sin(z) = \sin(x)\cdot\cosh(y) \pm i\cdot\cos(x)\cdot\sinh(y)$$

REZIDUUMI:

POLOVI = onda kada je nazivnik = 0

1) Ako je pol 1. reda u z_0

$$\mathrm{Rez}(f, z_0) = \lim_{z \to z_0} (z - z_0)\cdot f(z)$$

↳ $f(z)$ ima singularnost u z_0

↳ $\lim_{z \to z_0} (z - z_0)^n f(z) \neq 0$

$\lim_{z \to z_0} (z - z_0)^{n+1} f(z) = 0$

⇒ pomnožiti s $(z - z_0)^n$ i vidjeti koji je najmanji n na za koji više nema beskonačnosti.

2.) Ako je pol n-tog reda u z_0 $(n > 1)$

$$\mathrm{Rez}(f, z_0) = \frac{1}{(n-1)!}\cdot\lim_{z \to z_0} \frac{d^{n-1}}{d z^{n-1}}\left[(z - z_0)^n \cdot f(z)\right]$$

VEKTORSKA I. dio

$\nabla(r^n \cdot \vec{r})$ $\quad r = \|\vec{r}\|$, $\vec{r}$ = radij vekt. $\longrightarrow \nabla(r^n \cdot \vec{r}) = r^n \cdot \underbrace{\nabla \vec{r}}_{1} + \vec{r} \cdot \nabla r^n$

$$\nabla(r^n \cdot \vec{r}) = r^n + \vec{r} \cdot \nabla r^n \qquad \nabla r^n = n(n+1) r^{n-2}$$

$$\nabla \cdot (\vec{a} \cdot \vec{r}) = \vec{a} \qquad \nabla \cdot (\vec{a} \times \vec{r}) = \emptyset$$

$$\frac{\partial}{\partial \vec{s}} = \vec{\nabla} \cdot \vec{s} \longrightarrow (\varphi \nabla) \cdot \vec{f} = (\vec{\nabla} \varphi) \cdot \vec{f} = \nabla(\vec{\varphi} \vec{f})$$

* Primjer iz ispita: $\frac{\partial \vec{r}}{\partial \vec{a}} = \vec{\nabla}(\vec{a} \cdot \vec{r}) = \vec{a}$

! Usmjerena derivacija = MAT 2 $\quad \frac{\partial f}{\partial \vec{v}}(\vec{x_0}) = \nabla f \cdot \vec{v_0}$

$$\rightarrow \frac{\partial f}{\partial x}\Big|_{T_0} \cdot v_{01} + \frac{\partial f}{\partial y}\Big|_{T_0} \cdot v_{02} \qquad v_0 = \frac{\vec{v}}{\|\vec{v}\|}$$